

Multiple Choice

1. **Answer:** B

Options: A) 23.56 GHz; B) 42.58 MHz; C) 74.34 kHz; D) 13.93 MHz

Note: Correct option: B - 42.58 MHz

2. **Answer:** C

Options: A) 0.5 T; B) 1.2 T; C) 2.9 T; D) 100 T

Note: Correct option: C - 2.9 T

3. **Answer:** B

Options: A) The I⁻ : IO₃⁻ ratio is 3:1.; B) The MnO₄⁻ : I⁻ ratio is 6:5.; C) The MnO₄⁻ : Mn₂⁺ ratio is 3:1.; D) The H⁺ : I⁻ ratio is 2:1.

Note: Correct option: B - The MnO₄⁻ : I⁻ ratio is 6:5.

4. **Answer:** A

Options: A) hexane; B) 2-methylpentane; C) 3-methylpentane; D) 2,3-dimethylbutane

Note: Correct option: A - hexane

5. **Answer:** A

Options: A) MgO; B) MgS; C) NaF; D) NaCl

Note: Correct option: A - MgO

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.050 M AlCl₃'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'carbonic acid'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** I and III only. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 9 lines. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.